ARM PrimeCell

Smart Card Interface (PL131)

Release Note

PL131 RELN 0000A

Open Access

ARM PrimeCell Smart Card Interface (PL131) Release Note

Proprietary notice

Document confidentiality status

This document is Open Access. This document has no restriction on distribution.

Product status

The information in this document is Preliminary (information on a product in development).

Web address

http://www.arm.com

Feedback

ARM limited welcomes feedback on both the product, and the documentation.

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- The document title
- The documents number
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General suggestion for additions and improvements are also welcome.

Feedback on the ARM PrimeCell Smart Card Interface

If you have any comments or suggestions about this product, please contact your supplier giving:

- The Product name
- A concise explanation of your comments.

Change history

Issue	Date	Change
A	25th January, 2002	First release.

Contents

1 PRODUCT DELIVERABLES

1.1 ARM Part Numbers for this product

Table 1.1-1 lists the ARM part numbers for the individual deliverables that form part of the delivery for this ARM PrimeCell peripheral:

Product deliverables for the ARM PrimeCell PL131 peripheral	Document number	Product code
Synthesizable VHDL RTL	N/A	PL131-MN-23001-REL1v0
Synthesizable Verilog RTL	N/A	PL131-MN-22001-REL1v0
Synthesis Scripts (Synopsys Design Compiler, Avant! CB25 cell library)	N/A	PL131-MN-01002-REL1v0
Testbench - Functional Verification, VHDL	N/A	PL131-MN-23010-REL1v0
Testbench - EASY Integration test, VHDL	N/A	PL131-MN-23012-REL1v0
Testbench - Functional Verification, Verilog	N/A	PL131-MN-22010-REL1v0
Testbench - EASY Integration test, Verilog	N/A	PL131-MN-22012-REL1v0
Test vectors-BusTalk, functional verification	N/A	PL131-VE-01010-REL1v0
Test vectors-BusTalk, integration test	N/A	PL131-VE-01012-REL1v0
ARM PrimeCell Smart Card Interface (PL131) Technical Reference Manual (PDF)	ARM DDI 0228 A	PL131-DA-03001-REL1v0
ARM PrimeCell Smart Card Interface (PL131) Design Manual (PDF)	PL131 DDES 0000 A	PL131-DC-02002-REL1v0
ARM PrimeCell Smart Card Interface (PL131) Integration Manual (PDF)	PL131 INTM 0000 A	PL131-DC-10002-REL1v0
ARM PrimeCell Smart Card Interface (PL131) Release Note (PDF)	PL131 RELN 0000 A	PL131-DC-06002-REL1v0
ARM PrimeCell Smart Card Interface (PL131) User Guide (PDF)	PL131 USGD 0000 A	PL131-DC-08001-REL1v0
ARM PrimeCell Smart Card Interface (PL131) Errata document (PDF)	PL131 DEI 0000 A	PL131-DC-11001-REL1v0
System Common Source Code	N/A	PL000-SW-02001-REL7v0
System Indirection Source Code	N/A	PL000-SW-02003-REL7v0
System Test Source Code	N/A	PL000-SW-02004-REL7v0
System Build source code	N/A	PL000-SW-02005-REL7v0
Interrupt Controller Object Code	N/A	PL001-SW-00000-REL7v0
Timer Object Code	N/A	PL002-SW-02001-REL7v0
Software Code Module	N/A	PL131-SW-02001-REL2v0
Software Test Report	PL131_TREP_1000	PL131-DC-01001-REL2v0
Software API Document	PL131_ISPC_1000	PL131-DC-02001-REL2v0

System Structure and Integration Guide	PL000_DII_1000	PL000-DC-02000-REL7v0
System Common API Documents	PL000_ISPC_1001	PL000-DC-02001-REL7v0
System Indirection API Documents	PL000_ISPC_1002	PL000-DC-02001-REL7v0
System Test API Documents	PL000_ISPC_1003	PL000-DC-02003-REL7v0
System Common Files API Documents	PL000_ISPC_1004	PL000-DC-02004-REL7v0
Interrupt Controller API Documents	PL001_ISPC_1000	PL001-DC-02001-REL7v0

Table 1.1-1: ARM part numbers for ARM PrimeCell SCI PL131

Note Only the required deliverables will be supplied, as per the contractual agreement.

For each of the above, two files are generated using the UNIX tar create and the non-document tar files are then UNIX compress'ed. The files are named for each unique product revision, and the generic naming convention is illustrated below :

```
PL131-AA-NNNNN-REL1v0.tar.Z     - compressed tar file

PL131-AA-NNNNN-REL1v0.lst       - directory and file listing of above tar file
```

Where `-AA-NNNNN` represents the alpha-numeric product sub-type,
and REL1v0 represents the numeric release version of this peripheral.

1.2 File deliverables

This ARM PrimeCell peripheral consists of the following compressed 'tar' and 'list' file deliverables:

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/PL131-DC-06002-REL1v0.pdf

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/DAtasheet/PL131-DA-03001-REL1v0/PL131-DA-03001-REL1v0.lst
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/DAtasheet/PL131-DA-03001-REL1v0/PL131-DA-03001-REL1v0.tar

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/DoCumentation/PL131-DC-02002-REL1v0/PL131-DC-02002-REL1v0.lst
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/DoCumentation/PL131-DC-02002-REL1v0/PL131-DC-02002-REL1v0.tar

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/DoCumentation/PL131-DC-10002-REL1v0/PL131-DC-10002-REL1v0.lst
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/DoCumentation/PL131-DC-10002-REL1v0/PL131-DC-10002-REL1v0.tar

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/DoCumentation/PL131-DC-11001-REL1v0/PL131-DC-11001-REL1v0.lst
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/DoCumentation/PL131-DC-11001-REL1v0/PL131-DC-11001-REL1v0.tar

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/DoCumentation/PL131-DC-08001-REL1v0/PL131-DC-08001-REL1v0.lst *****
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/DoCumentation/PL131-DC-08001-REL1v0/PL131-DC-08001-REL1v0.tar *****

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-01002-REL1v0/PL131-MN-01002-REL1v0.lst
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-01002-REL1v0/PL131-MN-01002-REL1v0.tar.Z

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-22001-REL1v0/PL131-MN-22001-REL1v0.lst
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-22001-REL1v0/PL131-MN-22001-REL1v0.tar.Z

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-22010-REL1v0/PL131-MN-22010-REL1v0.lst
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-22010-REL1v0/PL131-MN-22010-REL1v0.tar.Z

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-22012-REL1v0/PL131-MN-22012-REL1v0.lst
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-22012-REL1v0/PL131-MN-22012-REL1v0.tar.Z

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-23001-REL1v0/PL131-MN-23001-REL1v0.lst
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-23001-REL1v0/PL131-MN-23001-REL1v0.tar.Z

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-23010-REL1v0/PL131-MN-23010-REL1v0.lst
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-23010-REL1v0/PL131-MN-23010-REL1v0.tar.Z

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-23012-REL1v0/PL131-MN-23012-REL1v0.lst
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/MN_native_model/PL131-MN-23012-REL1v0/PL131-MN-23012-REL1v0.tar.Z

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/VEctors/PL131-VE-01010-REL1v0/PL131-VE-01010-REL1v0.lst
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/VEctors/PL131-VE-01010-REL1v0/PL131-VE-01010-REL1v0.tar.Z

PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/VEctors/PL131-VE-01012-REL1v0/PL131-VE-01012-REL1v0.lst
PL131-BU-00000-REL1v0/PL131-BU-01000-REL1v0/VEctors/PL131-VE-01012-REL1v0/PL131-VE-01012-REL1v0.tar.Z

* A peripheral specific user guide document (PDF) is supplied. This describes instructions for synthesis and simulation.

(Note: Please ignore any generic user guide that may be supplied with this release).

An install script is supplied which uncompresses and tar extracts (un-tar) the compressed tar files in order to build the peripheral directory structure.

PLXXX_PrimeCell/PL000-BU-00003-REL2v0/SoftWare/PL000-SW-99000-REL2v0/INSTALL

A reference cell library (Avant! CB25 0.25µm standard cells) is supplied with this peripheral.

PLXXX_PrimeCell/PL000-BU-00003-REL2v0/Avanti/CB25_1999.1.tar.Z
PLXXX_PrimeCell/PL000-BU-00003-REL2v0/Avanti/CB25_1999.1.lst

1.2.1 Peripheral specific driver bundle

PLXXX_PrimeCell/PL131-BU-00000-REL1v3/PL131-BU-02000-REL2v0/Software/PL131-SW-02001-REL2v0/PL131-SW-02001-REL2v0.tar.Z
PLXXX_PrimeCell/PL131-BU-00000-REL1v3/PL131-BU-02000-REL2v0/Software/PL131-SW-02001-REL2v0/PL131-SW-02001-REL2v0.lst

PLXXX_PrimeCell/PL131-BU-00000-REL1v3/PL131-BU-02000-REL2v0/DoCumentation/PL131-DC-01001-REL2v0/PL131-DC-01001-REL2v0.tar
PLXXX_PrimeCell/PL131-BU-00000-REL1v3/PL131-BU-02000-REL2v0/DoCumentation/PL131-DC-01001-REL2v0/PL131-DC-01001-REL2v0.lst

PLXXX_PrimeCell/PL131-BU-00000-REL1v3/PL131-BU-02000-REL2v0/DoCumentation/PL131-DC-02001-REL2v0/PL131-DC-02001-REL2v0.tar
PLXXX_PrimeCell/PL131-BU-00000-REL1v3/PL131-BU-02000-REL2v0/DoCumentation/PL131-DC-02001-REL2v0/PL131-DC-02001-REL2v0.lst

1.2.2 Common driver bundle

PLXXX_PrimeCell/PL000-BU-00004-REL7v0/Software/PL000-SW-02001-REL7v0/PL000-SW-02001-REL7v0.tar.Z
PLXXX_PrimeCell/PL000-BU-00004-REL7v0/Software/PL000-SW-02001-REL7v0/PL000-SW-02001-REL7v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL7v0/Software/PL000-SW-02003-REL7v0/PL000-SW-02003-REL7v0.tar.Z
PLXXX_PrimeCell/PL000-BU-00004-REL7v0/Software/PL000-SW-02003-REL7v0/PL000-SW-02003-REL7v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL7v0/Software/PL000-SW-02004-REL7v0/PL000-SW-02004-REL7v0.tar.Z
PLXXX_PrimeCell/PL000-BU-00004-REL7v0/Software/PL000-SW-02004-REL7v0/PL000-SW-02004-REL7v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL7v0/Software/PL000-SW-02005-REL7v0/PL000-SW-02005-REL7v0.tar.Z
PLXXX_PrimeCell/PL000-BU-00004-REL7v0/Software/PL000-SW-02005-REL7v0/PL000-SW-02005-REL7v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL7v0/Software/PL001-SW-00000-REL7v0/PL001-SW-00000-REL7v0.tar.Z
PLXXX_PrimeCell/PL000-BU-00004-REL7v0/Software/PL001-SW-00000-REL7v0/PL001-SW-00000-REL7v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL7v0/Software/PL002-SW-02001-REL7v0/PL002-SW-02001-REL7v0.tar.Z
PLXXX_PrimeCell/PL000-BU-00004-REL7v0/Software/PL002-SW-02001-REL7v0/PL002-SW-02001-REL7v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL7v0/DoCumentation/PL000-DC-02001-REL7v0/PL000-DC-02001-REL7v0.tar
PLXXX_PrimeCell/PL000-BU-00004-REL7v0/DoCumentation/PL000-DC-02001-REL7v0/PL000-DC-02001-REL7v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL7v0/DoCumentation/PL000-DC-02003-REL7v0/PL000-DC-02003-REL7v0.tar
PLXXX_PrimeCell/PL000-BU-00004-REL7v0/DoCumentation/PL000-DC-02003-REL7v0/PL000-DC-02003-REL7v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL7v0/DoCumentation/PL000-DC-02004-REL7v0/PL000-DC-02004-REL7v0.tar
PLXXX_PrimeCell/PL000-BU-00004-REL7v0/DoCumentation/PL000-DC-02004-REL7v0/PL000-DC-02004-REL7v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL7v0/DoCumentation/PL000-DII-10000-REL7v0/PL000-DII-10000-REL7v0.tar
PLXXX_PrimeCell/PL000-BU-00004-REL7v0/DoCumentation/PL000-DII-10000-REL7v0/PL000-DII-10000-REL7v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL7v0/DoCumentation/PL001-DC-02001-REL7v0/PL001-DC-02001-REL7v0.tar
PLXXX_PrimeCell/PL000-BU-00004-REL7v0/DoCumentation/PL001-DC-02001-REL7v0/PL001-DC-02001-REL7v0.lst

2 INSTALLATION

2.1 Introduction

Intellectual Property (IP) deliverables are collectively delivered as a single UNIX compressed tar file which has been security encrypted.

An ARM PrimeCell peripheral may be one of several products delivered as part of this single encrypted file. Each ARM PrimeCell peripheral comprises a set of compressed tar files which correspond to a unique part number quoted in the contractual agreement document. A corresponding list file details the files in each tar file

These installation instructions cover the following platform/operating system:

- UNIX

2.2 Disk space required

This installation will require 25 Mbytes of free disk space.

2.3 Installation procedure

You will have a single tarred, zipped and encrypted file of following format:

e.g. Company Name-PrimeCell-PL131–Bundled-arm-download-WWWW.tgz.pgp

The installation procedure is summarised below:

2.3.1 Unpacking the shipment

The following steps describe how to unpack the separate products delivered in this shipment.

The first three steps may have already been done, ie the shipment from ARM may have already been decrypted and unzipped. If this is the case then go to step 4, else start with step 1.

1. **Relocate the shipment file**

   Relocate the encrypted file (for example <file>.tgz.des, if DES encryption has been used) to an appropriate location that has sufficient free disk space if PGP encrypted will need the relevant keys.

2. **Decrypt file**

   Follow the accompanying instructions for decryption of the file supplied, to generate a decrypted, compressed, tar file <file>.tgz

3. **Uncompress file**

   Execute the UNIX uncompress command to obtain the tar file <file>.tar:

   gunzip <file>.tgz

   where <file> is the name of the file supplied.

4. Extract tar files

To extract individual product tar files, execute the UNIX tar extract command:

```
tar -xvf <file>.tar
```

This will generate the following:

PL131-BU-01000-1.00 (hardware)

PL131-BU-01000-2.00 (software drivers)

ARM_MANIFEST_XXXX.TXT

ARM_DELIVERY_XXXX.TXT

that contains individual tar and list files for all of the products within this single shipment. Each peripheral consists of a set of tar files, and each tar file is named with the ARM part number that is listed in the contractual agreement document.

Note: It is suggested that part numbers delivered are checked against the contractual agreement document

2.3.2 Re-creating the peripheral directory structure

The following steps describe how to re-create the directory structure for each separate peripheral, which contains HDL RTL files, synthesis scripts, testbenches and documentation. These steps will need to be repeated for other peripherals in this shipment.

5. Create A Destination Directory

If installing multiple PrimeCells, it is recommended that they are each installed in SEPARATE directories such that each peripheral should ideally have its own destination directory.

Please ignore the following statement which is contained in the INSTALL script:

<You can install multiple PrimeCells from the same distribution into the same destination by specifying them on the command line.>

Create a unique destination directory for the PL131.

6. Method of installation

The directory structure should be extracted using the INSTALL script provided. The INSTALL script moves the compressed tar files delivered, to a user specified destination directory and then uncompresses and tar extracts the files to re-create the peripheral directory structure.

All files begin with PL131, where 131 is a 3-digit value unique to each peripheral.

Note: If the destination directory already exists, any files overwritten by this new distribution will be renamed to include a ".bak" extension.

7. Run the install script

The install script supplied will automatically uncompress and tar extract files for the specified peripheral revision (hardware files only), in a user specified directory. When step 6 is complete there is then no need to perform steps 7 and 8, instead go to step 9.

Change to the directory containing the INSTALL script, for example (note, revision may be later than –REL2v2)

```
cd PL000-SW-99000-2.02/
```

To invoke the install C-shell script supplied, enter the following at the UNIX command line:

```
./INSTALL <PRODUCT_NUMBER> <REVISION> <DESTINATION_DIRECTORY>
```

Options

<PRODUCT_NUMBER> = Each peripheral is defined by unique product code using the characters PL + 3-numeric digits. See below the shipment directory PL131_PrimeCell/

<REVISION> = Each peripheral has a revision number which has a status label and numeric version. See two directories below the shipment directory PL131_PrimeCell/

<DESTINATION_DIRECTORY> = the full directory path where the peripheral is to be installed.

For example

A product shipped with the directory structure /PL131-BU-00000-REL1v0/PL131-BU-00000-REL1v0/ has a product code PL131 and revision number Release REL1v0.

To install this peripheral to the the directory /ARM/PrimeCell_peripherals/PL131 enter the following command line

```
./INSTALL PL131 REL1v0 /ARM/PrimeCell_peripherals/PL131
```

This will install the peripheral having a product number PL131, with revision number REL131, to the destination directory /ARM/PrimeCell_peripherals/PL131

8. Peripheral directory installed

When all the peripheral files have been extracted, the destination directory (/ARM/PrimeCell_peripherals/) will contain:

- the top-level peripheral directory (e.g. sci_pl131) and underlying file structure of the deliverables, as well as the Release Note document file (PDF).
- The remaining documentation for the peripheral (in Adobe Portable Document Format, *.pdf), is located in the sci_pl131/docs/ directory.

2.3.3 Installation of the reference cell library

The reference cell library should be copied manually and tar extracted in the directory containing the PrimeCell peripherals. An example is shown below to copy the Avant! CB25 1999.1 cell library to the directory /ARM/PrimeCell_peripherals

```
cd /ARM/PrimeCell_peripherals
cp /PLXXX_PrimeCell/PL000_generic/Avanti/CB25_1999.1.tar.Z .
```

Execute the UNIX uncompress and tar extract commands as below:

```
uncompress CB25_1999.1.tar.Z
tar -xvf CB25_1999.1.tar
```

2.3.4 Installation of the Generic User Guide

For instructions on performing simulations and synthesis, refer to the generic peripheral user guide and the instructions for installing this are as follows. Else use the specific user guide if this is a new product (after January 2002).

The most recent revision of the generic user guide should be copied manually to a suitable location. An example is shown below to copy the generic user guide (revision REL1v0) to the directory /ARM/PrimeCell_peripherals

```
cd /ARM/PrimeCell_peripherals
cp /PLXXX_PrimeCell/PL000_generic/DoCumentation/PL000-DC-08001-REL1v0/PL000-DC-08001-REL1v0.tar .
```

To extract the document file (PDF), execute the UNIX tar extract command:

```
tar -xvf PL000-DC-08001-REL1v0.tar
```

3 DOCUMENTATION

3.1 Adobe PDF

Documents are supplied as “Adobe PDF” (Portable Document Format) files. These files are readable on most common computer platforms and operating systems using the appropriate file reader. A suitable file reader can be downloaded from the Adobe World Wide Web site at http://www.adobe.com. Select “Acrobat Reader" and download the reader for your computer platform/operating system.

3.2 Peripheral Documents Supplied

The following documents are supplied for the ARM PrimeCell Smart Card Interface PL131:
The documents files are located in the sci_pl131/docs/ directory.

ARM PrimeCell Smart Card Interface PL131
Technical Reference Manual ARM DDI 0228

The technical reference manual provides details for programming and interfacing to this ARM PrimeCell peripheral block.

ARM PrimeCell Smart Card Interface PL131
Integration Manual PL131 INTM 0000

The integration manual provides information required to integrate this ARM PrimeCell block within a typical industry standard ASIC design flow.

ARM PrimeCell Smart Card Interface PL131
Design Manual PL131 DDES 0000

The design manual describes the implementation details of this ARM PrimeCell block and the testbenches and test programs used for functional verification.

ARM PrimeCell Smart Card Interface PL131
Errata Notice PL131 DEI 0000

The errata notice describes details of any known issues with this ARM PrimeCell block and any workarounds for these issues. In most cases this document will be blank or may not be present.

ARM PrimeCell Smart Card Interface PL131
User Guide PL131 USGD 0000

The user guide describes the instructions on performing simulations and synthesis for this ARM PrimeCell block. This specific user guide will be present for newer products (after January 2002).

4 TRIAL TESTING

4.1 Product Maturity

This product is newly developed IP and approved for final release.
At the time of release, it has not been implemented in production silicon.

4.2 Technical data summary

The following is a summary of the key technical points from testing of the ARM PrimeCell Smart Card Interface PL131.

	VHDL RTL	Verilog RTL
Cell library	Avant! Passport 1999.1 CB25 v2.1 (0.25 µm)	Avant! Passport 1999.1 CB25 v2.1 (0.25 µm)
Total cell area, without scan, NAND-2 equivalents	11930	12448
Net interconnect area, without scan, NAND-2 equivalents	6490	6626
Total cell area, scan inserted,NAND-2 equivalents	15686	16352
Net interconnect area, scan inserted,NAND-2 equivalents	8876	9069
Clock frequency used	100 MHz	100 MHz
Scan insertion figure % (number of 'D' flip-flops replaced)	100% (1280)	100% (1280)
Estimated scan fault coverage, collapsed.	>99%	>99%
Synthesis and scan test tools and version	Synopsys Design Compiler and Test Compiler 2000.11-SP1	Synopsys Design Compiler and Test Compiler 2000.11-SP1
Simulation tools and versions	Model Technology Inc, ModelSim v5.5a	Model Technology Inc, ModelSim v5.5a Cadence LDV 3.2
Global Peripheral Development Environment Version	Release REL1v5	

4.3 BusTalk Functional block verification test scenarios

Functional block verification has been performed with the following scenarios using the BusTalk testbench with free-running PCLK.

Note Simulation log files are only supplied for RTL simulations for this beta release.

4.3.1 Synchronous clocks

PCLK, SCICLK	Simulations
100 MHz (10ns period)	VHDL and Verilog, RTL and gate-level min/max SDF

For further details of functional verification testing, refer to the ARM PrimeCell Smart Card Interface (PL131) Design Manual.

5 DIFFERENCES FROM PREVIOUS REVISIONS

Initial Release

6 KNOWN ISSUES

No known issues.

7 REVISION HISTORY

Date	Revision	Description
25-Jan-2002	REL1v0	Initial Release 1.0